Familiarity, More Than Stored Knowledge, Decides What Small Transformers Take from Context

Anonymous ACL submission

Abstract

Small transformers trained on text in which nearby sentences describe the same kind learn to carry a feature from one mention of a noun to the next. Probes of this use nonce nouns, but a nonce noun is both unfamiliar and without stored knowledge. We separated the two in a registered experiment. In 90 models (3.4M to 25.6M parameters, ten seeds) trained on an artificial language, 40 nonce nouns appeared 0 to 10,000 times: with one fixed shape, with a new shape every time, or never with any shape. In models trained on grouped text with stable shapes, carry of a shape given in the context fell from 4.5 nats for unseen nouns to under 1 nat by 100 exposures for every kind of noun, including nouns never seen with a shape. Stored knowledge of a fixed shape grew to about 10 nats but did not make carry fall faster; fixed-shape nouns carried only 0.3 to 0.8 nats less than shape-absent nouns at every exposure (exploratory). Familiar shape-absent nouns carried 0.33 nats more than unseen nouns given an embedding typical of trained nouns, a difference within a registered 1-nat equivalence bound. When training kept the context useful for familiar nouns, they kept more carry, and their history mattered more (exploratory). Ablating heads chosen by an induction-head signature did not selectively remove carry.

1 Introduction

A language model can take a word's properties from the text around it or from what it has stored about the word. Which it does depends on its training data. Classes that occur in bursts or rarely favour learning in context, and common classes favour learning stored in the weights (Chan et al., 2022, 2025); in pretrained models, heads that copy from the context compete with recall from memory (Yu et al., 2023; Jin et al., 2024). These results describe populations of items. We ask about a single word: as a model sees a noun more often, when does it stop taking the noun's features from the context, and why?

Two things change as a model sees a word. Its representation becomes like those of other words it knows, and it may store what the word's referents are like. In natural data the two go together, and probes inherit the confound: a nonce word tests generalisation to a new item (Thrush et al., 2020), but it is new in representation as well as in knowledge, and conclusions drawn with nonce words can depend on how the new word is represented (Kim et al., 2022; Cacioli, 2026). We met this confound in our own earlier experiments (Section 3). Small transformers trained on grouped text carried a shape from one mention of an unseen noun to the next, but most of that carry vanished when the same noun was given an embedding typical of trained nouns (an exploratory rescoring).

We therefore separated familiarity from stored knowledge by design. In every model, nonce nouns appeared a set number of times (0, 10, 100, 1,000 or 10,000) and, independently, either always with one shape, with a new shape every time, or never with any shape. The hypotheses and the analysis were registered before any corpus was built, with details fixed in a dated addendum before any data (Appendix M). We find:

1. **Familiarity did most of the work.** In grouped models with stable shapes, carry fell from 4.5 nats for unseen nouns to under 1 nat by 100 exposures for every kind of noun, including nouns never seen with a shape.

2. **Stored knowledge added a little, not a faster fall.** A fixed shape was learned strongly, but carry did not fall faster for it; fixed-shape nouns carried 0.3 to 0.8 nats less than shape-absent nouns throughout. Familiar shape-absent nouns ended close to unseen nouns given a typical embedding.

3. **When the context stayed useful, more carry**

remained and history mattered more. Where a familiar noun's shape could only come from the context, familiar nouns kept some carry, and less when their history gave a stored or an unpredictable shape (exploratory).

4. **We did not find the heads.** Heads chosen by an induction-head signature did not selectively remove carry.

Our claims are about transformers trained on an artificial language, not about children or large pretrained models.

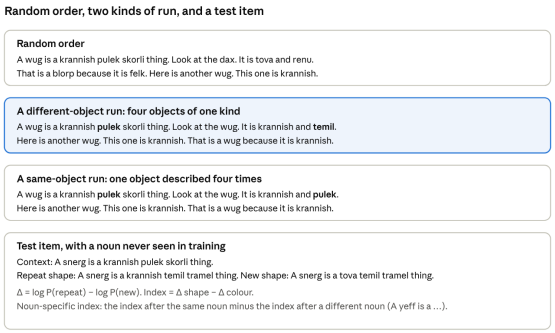


Figure 1: Random order, the two kinds of run used in earlier experiments, and a test item.

2 Related work

In-context and in-weights learning. In-context learning emerges when items occur in bursts, more when their meanings change, and trades off against learning stored in the weights (Chan et al., 2022). It favours rare classes and can fade as stored solutions take over (Chan et al., 2025; Singh et al., 2023), and its onset has been traced to induction heads (Olsson et al., 2022; Reddy, 2024). Closest to our question, periodically resetting embeddings during training keeps in-context use alive for rare and unseen tokens (Anand et al., 2025), and in synthetic biographies the most frequent entities shift from context to stored knowledge (Kim et al., 2026). In both, how familiar an item is goes together with whether a stored answer exists; we manipulate the two separately for single words within one model.

Context against memory. Pretrained models answer from memory more for popular or frequent entities and need retrieval for rare ones (Mallen et al., 2023; Kandpal et al., 2023); real entities are harder to move by context than invented ones (Du et al., 2024). Attention heads that copy from the context compete with mechanisms that recall stored facts, and removing a few heads can shift the balance (Yu et al., 2023; Jin et al., 2024). In this work frequency is observed rather than manipulated, and a familiar entity is also one the model knows about.

Nonce words and exposure. Nonce words are the usual test of generalisation to a new item (Thrush et al., 2020; Misra et al., 2023). Lexical exposure can inflate apparent generalisation (Kim et al., 2022), and controlling how often an item occurs in training shows how frequency shapes what transformers learn (Wei et al., 2021). We control exposure exactly, down to the window in which each occurrence is read.

Arrangement and repeated data. Packing related documents together changes how models use context (Shi et al., 2024; Zhao et al., 2024), and repeated passes over fixed data change what is learned (Murty et al., 2023; Charton and Kempe, 2024). Our earlier experiments vary only arrangement (Section 3).

3 Setting and earlier experiments

Language. Each corpus describes 32 kinds of object, each named by a nonce noun, with 14 exemplars per kind. Each exemplar has a shape, a colour and a texture, each a nonce word from a list of ten, and is described in one of five frames, such as *A wug is a zibby lavo shorfi thing* (Appendix I). In **Regular** corpora each kind keeps one shape. In **Reassigned** corpora a kind gets a new shape in every pass through the corpus and keeps it within the pass, so only the context can tell a familiar noun's current shape. In **random** order every pass is shuffled; in **grouped** order sentences about a kind come in runs of four (Figure 1). Each pass is arranged afresh and read in non-overlapping 128-token windows, so no text is read twice. Models are decoder-only transformers of 3.4M, 11.0M and 25.6M parameters (4 to 8 layers; Appendix C). Every artificial word is one token.

Earlier results. In registered experiments with 120 models (five seeds), we gave a model one sentence about an object of a new kind and measured how much more it expected the next mention of that noun to keep the shape than the colour (Appendices A and E). With nouns never seen in training, grouping raised this expectation over random order by 11.08 nats [9.57, 12.59], only when kinds had a fixed feature, and when Reassigned training made the context necessary, trained nouns carried the context's shape only after the same noun (Appendix L). The direction of these effects held when

Contrast	Unseen	Typical	Trained
Grouped − random	11.08	3.00	2.03
	[9.57, 12.59]	[1.79, 4.21]	[1.68, 2.37]
Regular − No stable, grouped	13.50	4.98	5.76
	[11.48, 15.52]	[3.42, 6.53]	[4.98, 6.53]
Different − same object	1.72	0.60	0.58
	[0.54, 2.90]	[0.26, 0.93]	[0.30, 0.86]

Table 1: Earlier study (index of shape over colour): seed-level differences in the shape-over-colour index (nats, 95% intervals over five seeds) for test nouns that were unseen, given a typical embedding, or trained. Registered for unseen nouns; the other columns are exploratory. No stable: no feature fixed within a kind.

History	Sentences about the noun in training
Fixed shape	always one shape, its own shape s^*
Uninformative	a new shape every time, drawn from the corpus's shapes
Shape-absent	only the sixth frame, which names no shape
Unseen	none (in the vocabulary only)

Table 2: Histories of the familiarity nouns. Each exposed history has four exposure levels (10 to 10,000).

the test nouns were given an embedding typical of trained nouns, and for trained nouns, but their size did not (exploratory; Appendix L). In grouped Regular models the preference to repeat the context's shape was 6.88 nats for unseen nouns and −0.72 for the same nouns with a typical embedding, and it fell steadily as we blended the two embeddings (-7.8 nats per unit blend). The unseen nouns' input embeddings had never been trained, because a tie between input and output embeddings was lost on the training hardware. So the earlier probes measured arrangement and the test noun's representation together. Models trained instead on one fixed file, read 160 to 290 times, memorised it, and their effects differed in size (Appendix C).

4 The familiarity study

Nouns. We added 40 nonce nouns to three base corpora (Regular grouped, Regular random and Reassigned grouped). In every model they fill 13 cells: three kinds of history (Table 2), each at 10, 100, 1,000 and 10,000 exposures, three nouns per cell, plus four unseen nouns. For comparison, a trained kind appears in about 60,000 sentences. Nouns rotate over cells across seeds. A sixth frame, *The noun is colour and texture*, names no shape; it is the only frame of shape-absent nouns and is also used by the trained kinds once in fifteen sentences, so it is not new.

Exact exposure. Each corpus holds exactly as many 128-token windows as training reads, and a fixed permutation says when each is read. Each familiarity sentence lies wholly inside one window, at most one per window and in grouped corpora only between runs, and a noun's exposures are evenly spaced from 5% to 95% of training. A checker re-read every corpus and verified each exposure, window and frame; the corpora matched checksums recorded before training. We trained 90 models: three sizes, ten seeds, three base corpora, with input and output embeddings untied in code.

Measures. Each item gives one context sentence, *A noun is a s colour texture thing*, then *A noun is a*, and reads the log probability L of each of the ten shapes at the next position. **Carry** is $L(s)$ minus the mean of L over shapes other than s and s^*; **stored preference** is $L(s^*)$ minus the same mean; **knowledge** is $L(s^*)$ minus the mean over the other nine shapes after an unrelated sentence. The context shape is never s^*, and each of the other nine shapes appears four times per noun. Unseen nouns are also scored with an embedding typical of trained nouns, as in the earlier study.

Hypotheses and analysis. In grouped Regular models we predicted that carry would fall faster with exposure for fixed-shape than for shape-absent nouns (H30); tested whether shape-absent nouns at 1,000 and 10,000 exposures differ in carry from unseen nouns with a typical embedding, with an equivalence test against ± 1 nat (H31); and predicted that stored preference rises with exposure (H32). A manipulation check (M1) required knowledge of at least 0.2 nats at 1,000 exposures, rising with exposure, in every base corpus. Head hypotheses (H33 to H35) are in Section 5.1. The seed is the unit of replication: we average cell means over the three sizes within each seed and test over ten seeds, with Holm (1979) correction across H30 to H35. A hypothesis is supported only if its direction also holds at two of the three sizes. A one-seed pilot, never used in a test, set the number of nouns per cell; its expected power for a 1-nat difference was 0.57, below the planned 0.8. The addendum fixed how a hypothesis counts as supported, the gate for H35 and how each test is computed (Appendix M).

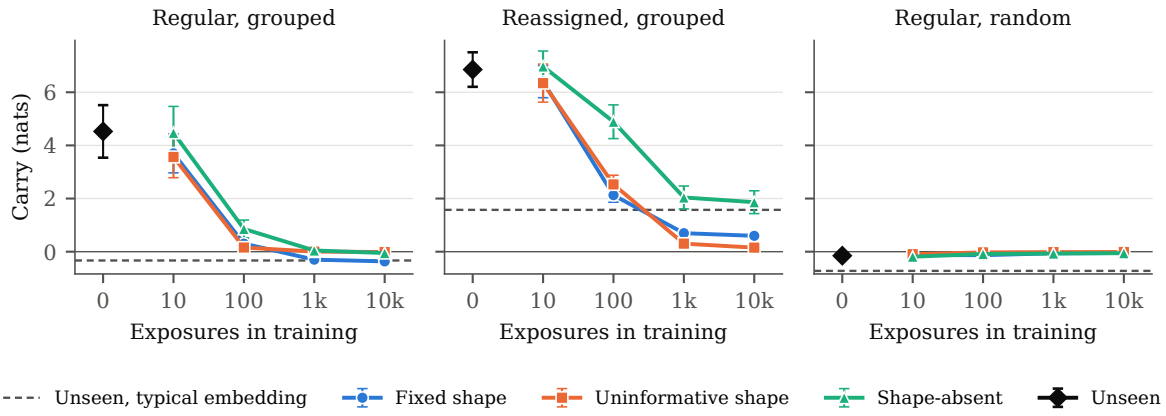


Figure 2: Carry of the context's shape by exposure and noun history (nats; means over ten seeds of size-averaged cell means, 95% intervals). Diamond: unseen nouns. Dashed line: unseen nouns with an embedding typical of trained nouns.

5 Results

The manipulation worked. Fixed-shape nouns learned their shape: knowledge at 1,000 exposures was 9.53, 8.61 and 8.26 nats in Regular grouped, Regular random and Reassigned grouped models, rising with exposure in all three (M1). In grouped Regular models, nouns with an uninformative or no shape showed no knowledge (within 0.2 nats; Table M3; descriptive).

Familiarity did most of the work. In grouped Regular models, carry was 4.53 nats [3.54, 5.52] for unseen nouns and had largely gone by 100 exposures for every history (Figure 2, left; Table M2). At 10 exposures it was 3.6 to 4.5 nats; at 100, 0.16 to 0.86; from 1,000 on, within 0.4 nats of zero. Grouping created the carry: grouped models carried more than random ones for unseen nouns (4.68 [3.71, 5.65]; R1, replicating the earlier study), and in random order carry was near zero for every noun (exploratory).

Stored knowledge did not speed the fall. H30 predicted that carry would fall faster with exposure for fixed-shape than for shape-absent nouns. The slope difference was 0.16 nats per decade [0.01, 0.30], in the opposite direction at every size, so H30 was not supported. Fixed-shape nouns started lower and stayed lower: they carried 0.77 [0.32, 1.22] nats less than shape-absent nouns at 10 exposures and 0.33 [0.25, 0.41] less at 1,000 and 10,000, where their carry was slightly negative (exploratory). Uninformative nouns carried less than shape-absent ones at 10 and 100 exposures (0.91 and 0.70 nats) but not from 1,000 on. Stored preference rose by 2.97 nats per decade [2.79, 3.16] for fixed-shape nouns (H32, Holm $p < .001$), reaching 10.71 nats at 10,000 exposures. So a stored shape lowered carry by a fraction of a nat, while familiarity lowered it by about four.

Familiar nouns came close to a typical embedding. H31 compared familiar shape-absent nouns (1,000 and 10,000 exposures) with unseen nouns given a typical embedding. The familiar nouns carried 0.33 nats [0.13, 0.52] more (Holm $p = .013$, positive at all three sizes), so H31 was supported as a difference, and the two were also equivalent within the registered 1-nat bound (TOST $p < .001$). Becoming familiar through training thus left a noun close to, but slightly above, where an ordinary-looking embedding puts it.

When the context stayed useful (exploratory). In Reassigned training a trained kind's current shape can come only from the context, and trained kinds carried it strongly (9.98 nats). Here familiarity reduced carry but did not remove it (Figure 2, middle). Shape-absent nouns kept 2.04 and 1.86 nats at 1,000 and 10,000 exposures, not reliably different from unseen nouns with a typical embedding (0.38 [−0.44, 1.19]; not shown equivalent). Fixed-shape nouns carried 1.31 nats [0.89, 1.72] less than shape-absent nouns, and uninformative nouns 0.42 [0.36, 0.47] less again (1,000 and 10,000 exposures pooled). History therefore mattered in both corpora, and more where familiarity left more carry to lower.

5.1 Heads

We registered a search for heads that carry the shape. In each model, before any familiarity item

was scored, we chose the k heads (k = 2, 4 or 6 by size) with the highest product of a prefix-matching score and a copying score on repeated random sequences of the language's words, the signature of induction heads (Olsson et al., 2022), and replaced their output by its mean over ordinary sentences at the position that predicts the shape. H33 predicted that this would remove more than half of the carry of unseen nouns in grouped Regular models, and more than ablating random heads of the same layers; H34 that it would reduce trained kinds' carry in Reassigned models; and H35, given H33, that the heads' attention to the context's shape would fall faster with exposure for fixed-shape nouns.

H33 was not supported. In grouped models the candidates were first-layer heads (all but one of 60 models), and the highest prefix-matching score in any model was 0.043, not clearly above the 0.027 of uniform attention (Table M4), so we describe them as copying heads, not induction heads. Ablating them removed a mean proportion of 0.22 [0.07, 0.36] of unseen carry per model, against the registered 0.5, and not reliably more than random heads did (difference 0.64 [–0.43, 1.71]). The proportion varied with size (exploratory): ablation removed all unseen carry in the 3.4M models (8.10 to –0.42 nats) but raised it in the 11.0M models (2.98 to 5.80), so the mean of proportions (0.22) is well below the proportion of mean carry removed (0.55). In the 3.4M and 11.0M models the first layer had exactly k non-candidate heads, so every random draw ablated the same set; the comparison with random heads rests mainly on that one set there. H34 was supported: in Reassigned models ablation cut trained kinds' carry by 7.41 nats [6.02, 8.80]. Ablating random heads cut it by 6.91 [5.45, 8.38] (exploratory), so the effect is not specific to the candidates. H35 was not tested. Exploratory measures of the candidates' attention to the context's shape fell with familiarity in grouped Regular models, from 0.155 for unseen nouns to about 0.07 from 1,000 exposures for every history, in parallel with carry (Figure 3; Table M5); in Reassigned models attention and carry did not order the histories the same way. Our registered ablations do not show that any specific heads produce the carry.

6 Discussion

In these models a noun took far less of its shape from the context once it was familiar, and familiarity did most of the work: a familiar noun that never appeared with a shape lost about as much carry as one with a fixed shape, and ended close to an unseen noun given an ordinary-looking embedding. A stored shape lowered carry by a further fraction of a nat but did not make it fall faster. One reading, which our measures did not test, is that the large carry measured with unseen nouns in the earlier study was carry for a token unlike any trained noun, and that it ends when the token's representation becomes ordinary, by training or by assignment. For single words in our language, this refines the trade-off between in-context and in-weights learning (Chan et al., 2022, 2025): most of the shift away from the context came with familiarity, not with a stored answer competing against the context. It fits the finding that keeping embeddings untrained preserves in-context use for rare and unseen tokens (Anand et al., 2025).

Where training kept the context useful for familiar nouns, more carry remained and a noun's history mattered more: a stored shape lowered carry, and so did a shape that had changed in every sentence. The second effect may be a noun-specific lesson about how a feature varies, loosely analogous to what hierarchical models call an overhypothesis (Kemp et al., 2007); it is exploratory, and we did not test that account.

For work with nonce words, an unseen token is not a neutral stand-in for a new word: its untrained or unusual representation can carry much of an effect, and effects should be checked with representations that look familiar. For mechanistic work, an induction-head signature did not find components that selectively carry a kind's feature in our models. First-layer attention to the shape word co-varied with carry in grouped Regular models, but a different, causal search would be needed to say what produces it.

Limitations

The language is artificial, with 32 kinds, three dimensions and six frames; the models are small (3.4M to 25.6M parameters) and trained with one recipe. We do not know whether the familiarity effect holds in pretrained models. H30 tested whether stored knowledge speeds the fall in carry; it did not, and the smaller effects of knowledge on the level of carry are exploratory. Exposure levels were spaced by factors of ten, and carry fell between 10 and 100 exposures, which we did not sample. Power for a 1-nat difference was below plan (0.57). Unseen

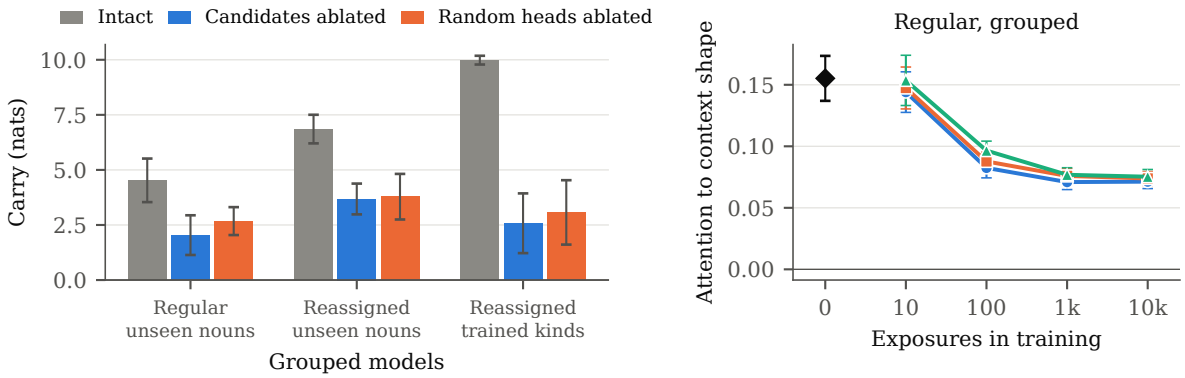


Figure 3: Left: carry intact, with the candidate heads mean-ablated, and with random heads ablated (nats; mean over ten seeds and three sizes, 95% intervals over seeds). Right: the candidates' attention from the position that predicts the shape to the context's shape word, grouped Regular models (exploratory).

nouns in the earlier study had untrained embeddings because an embedding tie was lost; in the familiarity study embeddings were untied by design. The head search used one registered signature and mean ablation; other methods might find components we missed. The earlier experiments were registered in stages, some after related results were known (Appendix A); the familiarity study was registered before any corpus was built, with details fixed in a dated addendum before data and a pilot that set only the number of nouns per cell.

Ethics statement

The corpus analysis in Appendix J is a secondary analysis of archived, de-identified transcripts (CHILDES, CABank) used under their terms; no data were collected from people, and no transcript excerpts are reproduced. Code, corpora, results and all trained models will be released; anonymised copies are provided as supplementary material.

Use of AI assistants

An AI assistant (Claude, Anthropic) was used to help design analyses, write code, review code and design, draft and edit text and check citations, and as the second coder in a consistency check of the corpus coding. The authors are responsible for all content, including the accuracy of every citation.

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A Registration record

In the appendices and the registrations, Study 1 is the training experiments (Section 3 and Appendices A to I), Study 2 the natural corpora (Appendix J) and Study 3 the pretrained models (Appendix K).

All registrations are in one public project on the Open Science Framework (link withheld for review; an anonymised view-only link is in the supplementary material), and each states what was known when it was written. Table A1 lists them, Table A2 gives the label each registration uses for each row of Table A4, and Table A4 gives every registered hypothesis and its result.

Table A1: Registrations. Each was written before the models it concerns were trained or scored, or before the transcripts were read.

Registration	Written	Rows of Table A4	Known when it was written
Original	2 Mar 2026	Corpora, models and controls (Appendix D)	Nothing trained
Amendment 1	7 Oct 2026	1, 2, 3, 6, 16	Random-order models trained and scored on an earlier test, with the old loader; no grouped model trained
Amendment 2	7 Oct 2026	7 (texture added as a co-primary test)	No grouped model trained
Amendment 3	8 Oct 2026	23, 24, 25	Study 1 results for rows 1 to 3, 6, 7 and 16 (old loader); no pretrained model scored
Amendment 4	8 Oct 2026	20, 21	Studies 1 and 3 results; no transcript read
Amendment 5	9 Oct 2026	4, 5, 8, 9, 17, 18, 19, 22	All earlier results (Study 1 with the old loader); four reviews of a draft; five exploratory checks, including the confound of the noun with another
Amendment 6	11 Oct 2026	11, 12, 13	Amendment 5 results (old loader); a prediction for the new models fitted to them
Amendment 7	12 Oct 2026	10, 14, 15	All results, with corrected scores
Amendment 8	15 Oct 2026	26 to 32	All results, with corrected scores; three rounds of reviews, which noted that fixed-file models memorised their files and that the novel-noun test was uninformative; no fresh-text model trained

Table A2: Rows of Table A4 and the labels used in the registrations.

Row	Label	Row	Label	Row	Label	Row	Label
1	H4b	9	H13b	17	H14a	25	H8
2	H1b	10	H22	18	H14b	26	H23
3	H2b	11	H17	19	H15	27	H24
4	H11	12	H18	20	H9	28	H25
5	H12	13	H19	21	H10	29	H26
6	H5b colour	14	H20	22	H16	30	H27
7	H5b texture	15	H21	23	H6	31	H28
8	H13a	16	H3b	24	H7	32	H29

The loading correction. The model code ties the input embedding to the output layer. On the graphics card used for training (through DirectML) the tie was lost, so every Study 1 model was trained and saved with two separate matrices. The first version of the scoring code rebuilt each model with the tie in place before loading its weights, so the output matrix overwrote the input embedding. We found this on 12 October 2026, fixed the loader, and rescored all 315 Study 1 models then trained, 461 intermediate checkpoints of the random and grouped Regular models and the trained-noun items, with the same items. A check on eight randomly chosen models confirmed the fix: mean loss on each model's own training

corpus was 0.07 to 0.16 nats per token with the corrected loader, close to the last logged training loss (0.08 to 0.23; dropout is off at test), and 0.58 to 1.97 with the old one. Studies 2 and 3 were not affected. The old scores are kept in the repository.

Table A3: Study 1 registered tests before and after the correction (shape-over-colour index, nats). Rows 10, 14 and 15 were registered and scored after the correction.

Row	Old loader	Corrected	Verdict before → after
1	1.36 vs 0.18	6.49 vs 0.47	Supported → supported
2	0.18	0.47	Not supported → not supported
3	0.18 vs 0.09	0.47 vs −0.25	Not supported → supported
4	difference 1.45	difference 6.93	Supported → supported
5	difference 1.17	difference 2.76	Supported → supported
6	1.36 vs 0.80	6.49 vs 2.12	Supported → supported
7	1.10 vs −0.04	5.36 vs 0.46	Supported → supported
8	slope 0.29	slope 1.63	Supported → supported
9	0.24 vs 0.18	1.43 vs 0.47	Not supported → supported
11	0.21 vs 0.18	0.50 vs 0.47	Not supported → not supported
12	0.29 vs 0.24	0.80 vs 1.43	Not supported → not supported (reversed)
13	0.21 vs −0.08	0.50 vs 0.10	Supported → supported
16	difference 0.0007	difference 0.005	Equivalent to zero → equivalent to zero

Table A4
All registered hypotheses and results; registered secondary analyses are reported in the text. Rows 1 to 16 concern fixed-file models and rows 26 to 32 fresh-text models. Seed-level tests average the three sizes within each of five seeds (t-based 95% intervals, two-sided; tests one-sided unless marked, 4 degrees of freedom), so a one-sided test can be below .05 while the two-sided interval includes zero (row 13). The registered p is the test named in the registration, one-sided unless marked, Holm-corrected within its registration; for rows 26 to 32 it is the seed-level test, corrected within rows 26 to 29 and within rows 30 to 32. Corpus intervals resample sources and then documents; Pythia and BabyLM intervals resample nouns.

#	Prediction	Result (nats)	Seed-level difference [95% CI]	Seed-level p	Registered p	Verdict
	1. Does grouping teach the fixed feature?					
1	Grouped order gives a higher index than random order	6.49 vs 0.47	6.02 [5.24, 6.79]	$< .001$	$< .001$	Supported
2	Random order gives an index above zero	0.47	0.47 [−0.61, 1.56]	.15	.061	Not supported
3	Random order: Regular above Mixed-slot	0.47 vs −0.25	0.73 [−0.69, 2.15]	.11	.032	Supported by the registered test, not at the seed level
4	Grouped: Regular above No stable feature	6.49 vs −0.44	6.93 [5.47, 8.40]	$< .001$	$< .001$	Supported
5	Grouped: Colour stable below No stable feature	−3.20 vs −0.44	2.76 [1.99, 3.52]	$< .001$	$< .001$	Supported
6	Grouped: Regular above Mixed-slot, shape over colour	6.49 vs 2.12	4.37 [1.98, 6.76]	.004	$< .001$	Supported
7	Grouped: Regular above Mixed-slot, shape over texture	5.36 vs 0.46	4.89 [2.69, 7.10]	.002	$< .001$	Supported
8	The index rises with log B across six orders	Slope 1.63 (SE 0.10)	1.63 [1.46, 1.80]	$< .001$	$< .001$	Supported
9	B = 3.6 above random order	1.43 vs 0.47	0.96 [0.63, 1.28]	$< .001$	$< .001$	Supported
10	The same, Objects corpora, different-object runs	1.18 vs 0.22	0.96 [0.37, 1.54]	.005	$< .001$	Supported
26	Fresh text: grouped order gives a higher index than random order	9.52 vs −1.56	11.08 [9.57, 12.59]	$< .001$	$< .001$	Supported
27	Fresh text: B = 3.5 above random order	1.28 vs −1.56	2.84 [1.41, 4.27]	.003	.005	Supported
28	Fresh text, grouped: Regular above No stable feature	9.52 vs −3.98	13.50 [11.48, 15.52]	$< .001$	$< .001$	Supported
	2. What kind of grouping helps?					
11	512-token window above 128, random order	0.50 vs 0.47	0.03 [−0.44, 0.50]	.44	.87	Not supported
12	512-token window above 128, B = 3.6	0.80 vs 1.43	−0.63 [−0.90, −0.37]	.003 (two-sided)	1.00	Not supported; reliably reversed
13	512-token window: Regular above No stable feature	0.50 vs 0.10	0.40 [−0.04, 0.85]	.032	.045	Supported; weak
14	Grouped: different-object above same-object runs	5.65 vs 2.99	2.66 [1.57, 3.75]	.001	$< .001$	Supported
15	B = 3.6: different-object above same-object runs	1.18 vs 0.45	0.73 [0.15, 1.32]	.013	$< .001$	Supported
29	Fresh text, grouped: different-object above same-object runs	7.60 vs 5.87	1.72 [0.54, 2.90]	.008	.008	Supported
	3. Is the expectation tied to the noun?					
16	Fixed files, random order: index higher after the same noun (novel nouns)	Difference 0.005; equivalent to zero ($p = .001$), but the novel nouns were untrained (Section 3)			.46	Not supported
30	Shape reassigned, grouped: noun-specific index above zero, novel nouns	0.37	0.37 [0.24, 0.49]	$< .001$	.001	Supported

#	Prediction	Result (nats)	Seed-level difference [95% CI]	Seed-level p	Registered p	Verdict
31	The same, above grouped Regular (fresh text)	0.37 vs 0.12	0.25 [0.10, 0.39]	.005	.005	Supported
32	Shape reassigned, grouped: noun-specific index above zero, nouns from training	11.24	11.24 [10.44, 12.03]	$< .001$	$< .001$	Supported
17	Pythia: noun-specific index rises with size (wording matched)	$\rho = -.42$			1.00	Not supported
18	Pythia 410M to 12B: noun-specific index above zero (wording matched)	−0.15 (SE 0.12)			1.00	Not supported
19	Pythia: 410M above 160M across ten training seeds 4. What do natural input and pretrained models show?	0.15 vs −0.15			.012	Supported
20	Speech to children: B above 1	3.57 [3.13, 4.30]			$< .001$	Supported
21	Speech to children above written text	3.57 vs 1.43; difference 2.14 [1.55, 2.91]			$< .001$	Supported
22	Speech to children differs from adult conversation (two-sided)	3.57 vs 2.33; difference 1.24 [0.51, 2.03]			$< .001$	Supported; higher
23	Pythia: index rises with size	$\rho = .92$			.003	Supported
24	Pythia: index rises over training	Mean within-size $\rho = -.34$			1.00	Not supported
25	BabyLM baselines (10M, 100M words): index above zero	0.47 [−1.05, 1.93]; 0.37 [−0.04, 0.73]			.52; .12	Not supported

Fresh text and shape reassignment. Rows 26 to 32 were registered on 15 October 2026, after reviews of the manuscript had noted that fixed-file models memorised their files and that the test with novel nouns was uninformative, and before any fresh-text model was trained. The corpora were built, and their checksums registered, before training. Amendment 8 names the conditions by their folder names (for example reassigned_grouped).

B Deviations from the registrations

1. Two registered analyses are not reported here: the Bayesian reference model of Amendment 1, which is in the repository, and the prediction of Amendment 6 for the 512-token models, which had been fitted to scores later corrected (Appendix A).

2. Added after registration: the seed-level tests for rows 1 to 3 and 6 to 8, the seed-level intervals, and intervals for the differences between natural corpora (from the registered two-stage bootstrap). Effective clustering was computed before it was registered as the scale for Figure I1 and is reported as exploratory (Table J1).

3. The two models of about 70 billion parameters were scored with Apple's MLX library, because PyTorch crashed the machine; on smaller models the two differed by 0.003 to 0.010 nats on average.

4. In the hand coding of Study 2, a rule for labels not used for a physical object was added during coding and applied by both coders, and one pair of the reliability subset was discussed with an AI assistant and left out of the agreement check.

Smaller technical changes (software versions, download addresses, a stale file in one public model repository) are listed in the repository.

C Model and training settings

Table C1: AdamW (betas 0.9 and 0.95, weight decay 0.1), linear warm-up then cosine decay, dropout 0.1, untied input and output embeddings. Fixed-file models read windows starting at random positions in the corpus; fresh-text models read non-overlapping windows, each at most once, in random order (the 3.4M and 25.6M models used about 60% of the windows available, the 11.0M models 94 to 96%). Byte-pair tokeniser with 351 entries for the Regular corpora and 314 to 339 for the other corpus families; every artificial word is one token. Seeds 42, 123, 456, 789 and 1001 each set the corpus, the initial weights and the batch order. Checkpoints were saved every 500 steps. Training ran on one Windows PC through DirectML.

Setting	3.4M	11.0M	25.6M
Layers, heads, width	4, 4, 256	6, 8, 384	8, 8, 512
Feed-forward width	1,024	1,536	2,048
Steps	5,000	8,000	10,000
Batch (128-token windows)	32	32	16
Batch (512-token windows)	8	8	4
Peak learning rate	3×10^{-4}	2×10^{-4}	1×10^{-4}
Warm-up steps	200	400	500

Table C2: Memorisation. Mean next-token loss (nats per token) on 300 windows from each model's own training text and from a new arrangement of the same sentences (100 further passes, 27 for the Objects corpora, never used in training); 15 models per cell. At the seed level, the gap between the two losses was larger for fixed-file than for fresh-text models by 0.57 to 1.16 nats, with every interval excluding zero.

Corpus	Fixed file: own file	Fixed file: new arrangement	Fresh text: own text	Fresh text: new arrangement
Regular, random	0.11	1.25	0.55	0.55
Regular, B 3.6 (fresh: 3.5)	0.10	1.26	0.54	0.54
Regular, grouped	0.09	0.85	0.35	0.35
No stable feature, grouped	0.09	1.00	0.39	0.39
Objects, same-object runs, grouped	0.08	0.66	0.28	0.28
Objects, different-object runs, grouped	0.09	0.76	0.32	0.32

D Further content controls

Six further corpora, each trained in random order only, change one thing about Regular. **No kind label**: the nouns are removed, with no other marker of the kind. **Paraphrased**: the nouns are removed and replaced by randomly chosen category markers. **Weak label**: nouns appear in 25% of sentences and category markers in the rest. **Frequency matched**: the same feature words as often as in Regular, with the fixed feature of each kind drawn at random from shape, texture or none. **Added noise**: Regular with 7% of exemplars given a wrong fixed feature. **Feature swap**: 16 kinds keep their shape and 16 keep their texture.

Table D1: Further controls, random order, 15 models each. In random order every index is small, so these corpora say little about what grouping would do with them.

Corpus	Shape over colour, mean (SD)	Shape over texture	Noun-specific index
Regular (for comparison)	0.47 (0.90)	0.51	0.005
No kind label	0.06 (0.49)	0.44	−0.004
Paraphrased	0.41 (0.44)	0.60	−0.025
Weak label	0.09 (0.50)	−0.52	0.008
Frequency matched	−0.22 (0.63)	−0.40	−0.033
Added noise	−0.43 (0.67)	0.44	−0.038
Feature swap	0.69 (0.77)	0.60	−0.017

E Robustness

Table E1
p values for the Study 1 rows of Table A4 under paired tests at the seed level and over the 15 matched models (one-sided unless marked).

Row	Seed level (5 seeds)	Matched models (15)
1	$< .001$	$< .001$
2	.15	.031
3	.11	.014
4	$< .001$	$< .001$
5	$< .001$	$< .001$
6	.004	$< .001$
7	.002	$< .001$
8	$< .001$	
9	$< .001$	$< .001$
10	.005	$< .001$
11	.44	.44
12	.003 (two-sided)	.003 (two-sided)
13	.032	.015
14	.001	$< .001$
15	.013	$< .001$
26	$< .001$	$< .001$
27	.003	$< .001$
28	$< .001$	$< .001$
29	.008	$< .001$
30	$< .001$	$< .001$
31	.005	.010
32	$< .001$	$< .001$

Table E2
Shape-over-colour index by model size (one exemplar), and with two or four exemplars in the context that share their shape (all sizes). Rows marked fresh are fresh-text models; the rest are fixed-file models. In grouped Regular models, two or four exemplars lowered the index: a run of a kind has four sentences, so a fifth sentence about the same kind is rare (Section 3). In grouped No stable feature and Colour stable models, where shape varies within a kind, more exemplars sharing a shape lowered the index (Δ shape −0.64, −1.93 and −3.42 with one, two and four exemplars in grouped No stable feature models), for which we have no tested account; in random-order Regular models, too, the index was lower with two or four exemplars than with one.

Condition	3.4M	11.0M	25.6M	Two exemplars	Four exemplars
Regular, random, fresh	−1.00	−0.87	−2.81	−1.83	−2.19
Regular, B 3.5, fresh	1.37	2.27	0.21	2.05	−0.53
Regular, grouped, fresh	7.43	12.57	8.56	7.88	4.24
No stable feature, grouped, fresh	−1.21	−4.80	−5.92	−5.41	−7.24
Objects, same-object, grouped, fresh	8.21	4.60	4.81	3.86	0.62
Objects, different-object, grouped, fresh	9.96	6.29	6.54	5.78	2.13
Reassigned, random, fresh	−0.06	0.33	0.35	0.59	0.23
Reassigned, grouped, fresh	6.52	14.05	8.85	9.62	8.26
Regular, random	0.33	0.71	0.38	−0.64	−0.10
Regular, B 1.4	0.73	1.05	0.70	−0.01	0.32
Regular, B 3.6	0.97	2.05	1.27	0.60	1.06
Regular, B 7	1.85	2.89	1.45	1.23	1.15
Regular, B 12	3.18	4.53	2.88	2.85	2.43
Regular, grouped	6.41	6.83	6.23	5.24	4.30
Mixed-slot, random	−0.11	−0.45	−0.21	0.29	0.29
Mixed-slot, grouped	2.24	2.35	1.76	2.60	2.04
No stable feature, random	−0.10	−0.33	−0.12	−0.17	−0.44
No stable feature, grouped	−0.41	−0.28	−0.63	−1.77	−3.26
Colour stable, random	−0.04	−0.70	−0.51	−0.76	−0.76
Colour stable, grouped	−2.11	−3.72	−3.77	−4.50	−5.34
Regular, random, 512	0.30	0.51	0.70	0.51	0.32
Regular, B 3.6, 512	0.31	1.02	1.06	0.93	0.82
No stable feature, random, 512	0.21	0.04	0.05	−0.10	−0.23
Objects, random	0.36	0.44	−0.13	0.38	1.02
Objects, same-object, B 3.6	0.22	0.98	0.14	0.32	1.22
Objects, different-object, B 3.6	0.83	1.75	0.96	1.49	2.18
Objects, same-object, grouped	3.00	3.33	2.62	1.30	1.34
Objects, different-object, grouped	6.18	5.43	5.34	4.54	4.14

Noun-specific index. At the seed level, the noun-specific index was equivalent to zero (two one-sided tests, ±0.2 nats) in 25 of the 26 fixed-file conditions; the exception was same-object runs at B = 3.6 (−0.11, p = .058). No condition exceeded 0.15 nats in absolute value. These nulls are uninformative, because the models did not track novel nouns (Section 3). As a further check, for the 60 models in the random and grouped Regular and grouped Objects conditions we compared the log probability of the same novel noun after the context and the word A with the mean over the other seven novel nouns. The difference was no more than 0.032 nats in absolute value in every model, and positive on about half the items. Because the novel nouns were never training targets, this check also depends on output weights that training only pushed down, so it is weaker evidence than the trained-noun contrast in Section 3.5.

Correction across all tests. Holm correction was applied within each registered family of tests, as registered. Every row of Table A4 with a Holm-adjusted p below .001 has an unadjusted p below .001, and so stays below .05 under Holm correction across all 32 tests.

What the corpora reward. For every pair of consecutive sentences in each training corpus (five seeds pooled), Table E3 gives the share about the same kind and how often the second sentence repeats the first one's shape or colour. The noun-blind reference index is D(shape) − D(colour), where D = ln p − ln((1 − p)/9) and p is the share of pairs that repeat that feature, with ten values per dimension; it is what a learner that predicts each feature from the previous sentence alone would show on a one-exemplar item. The noun-aware reference computes the same index separately for pairs about the same kind and pairs about different kinds (add-0.5 smoothing).

Table E3: What each training corpus says about the next sentence, and the models' shape-over-colour index for comparison. Rows marked fresh describe fresh text, counted on 100 held-out passes arranged by the same rule (27 for the Objects corpora); the rest describe the fixed files. All values in nats except the shares. Code is in the repository.

Corpus	Next sentence same kind	Shape repeats, same kind	Shape repeats, other kind	Colour repeats, same kind	Noun-blind reference	Noun-aware, same kind	Noun-aware, other kind	Model index
Regular, random, fresh	0.03	1.00	0.07	0.10	−0.01	11.7	−0.35	−1.56
Regular, B 3.5, fresh	0.11	1.00	0.07	0.09	0.65	13.1	−0.37	1.28
Regular, grouped, fresh	0.72	1.00	0.07	0.10	3.25	14.9	−0.38	9.52
No stable feature, grouped, fresh	0.72	0.10	0.10	0.10	0.02	0.0	−0.01	−3.98
Objects, same-object, grouped, fresh	0.76	1.00	0.07	0.99	0.39	8.4	−0.35	5.87
Objects, different-object, grouped, fresh	0.76	1.00	0.07	0.10	3.43	14.9	−0.29	7.60
Reassigned, random, fresh	0.03	1.00	0.10	0.10	0.27	8.2	0.01	0.21
Reassigned, grouped, fresh	0.72	1.00	0.10	0.10	3.30	11.4	−0.02	9.80
Regular, random	0.03	1.00	0.07	0.10	−0.01	10.3	−0.35	0.47
Regular, B 1.4	0.04	1.00	0.07	0.10	0.13	10.8	−0.35	0.83
Regular, B 3.6	0.11	1.00	0.07	0.10	0.70	11.7	−0.31	1.43
Regular, B 7	0.22	1.00	0.07	0.10	1.23	12.4	−0.37	2.06
Regular, B 12	0.38	1.00	0.07	0.09	1.91	13.0	−0.40	3.53
Regular, grouped	0.72	1.00	0.07	0.10	3.25	13.5	−0.33	6.49
No stable feature, grouped	0.72	0.10	0.10	0.10	0.03	0.0	0.06	−0.44
Colour stable, grouped	0.72	0.10	0.10	1.00	−3.26	−12.5	0.38	−3.20
Objects, same-object, B 3.6	0.11	1.00	0.07	0.70	0.19	8.6	−0.35	0.45
Objects, different-object, B 3.6	0.11	1.00	0.07	0.11	0.74	11.6	−0.25	1.18
Objects, same-object, grouped	0.76	1.00	0.07	0.99	0.37	7.0	−0.39	2.99
Objects, different-object, grouped	0.76	1.00	0.07	0.10	3.39	13.5	−0.41	5.65

Table E4: Study 1 results for models trained on fixed files, in nats after one exemplar with the same noun; 15 models per row, 128-token windows unless marked.

Corpus	Order (B)	Δ shape	Δ colour	Shape over colour, mean (SD)	Models above zero	Shape over texture
Regular	Random (0.9)	0.50	0.03	0.47 (0.90)	10 of 15	0.51
Regular	Graded (1.4)	0.77	−0.05	0.83 (0.72)	12	0.81
Regular	Graded (3.6)	1.33	−0.10	1.43 (0.93)	15	1.28
Regular	Graded (7)	1.71	−0.36	2.06 (0.95)	15	1.95
Regular	Graded (12)	2.94	−0.59	3.53 (1.12)	15	3.09
Regular	Grouped (23)	5.36	−1.13	6.49 (1.49)	15	5.36
Mixed-slot	Random	−0.37	−0.12	−0.25 (0.72)	4	−0.54
Mixed-slot	Grouped	0.76	−1.36	2.12 (1.07)	14	0.46
No stable feature	Random	−0.01	0.17	−0.18 (0.39)	5	−0.09
No stable feature	Grouped	−0.64	−0.20	−0.44 (0.54)	4	−1.02
Colour stable	Random	−0.06	0.36	−0.42 (0.70)	4	−0.06
Colour stable	Grouped	−0.81	2.39	−3.20 (1.18)	0	−0.22
Regular	Random (0.9), 512-token window	0.37	−0.13	0.50 (0.62)	12	0.41
Regular	Graded (3.6), 512-token window	0.67	−0.13	0.80 (0.79)	11	0.67
No stable feature	Random, 512-token window	0.07	−0.03	0.10 (0.60)	9	−0.10
Objects	Random (1.0)	0.58	0.36	0.22 (0.43)	11	0.40
Objects	Same-object runs (3.6)	1.34	0.90	0.45 (0.62)	12	1.40
Objects	Different-object runs (3.6)	1.25	0.07	1.18 (0.67)	14	1.13
Objects	Same-object runs (24)	4.63	1.65	2.99 (1.05)	15	4.05
Objects	Different-object runs (24)	5.37	−0.28	5.65 (1.18)	15	5.07

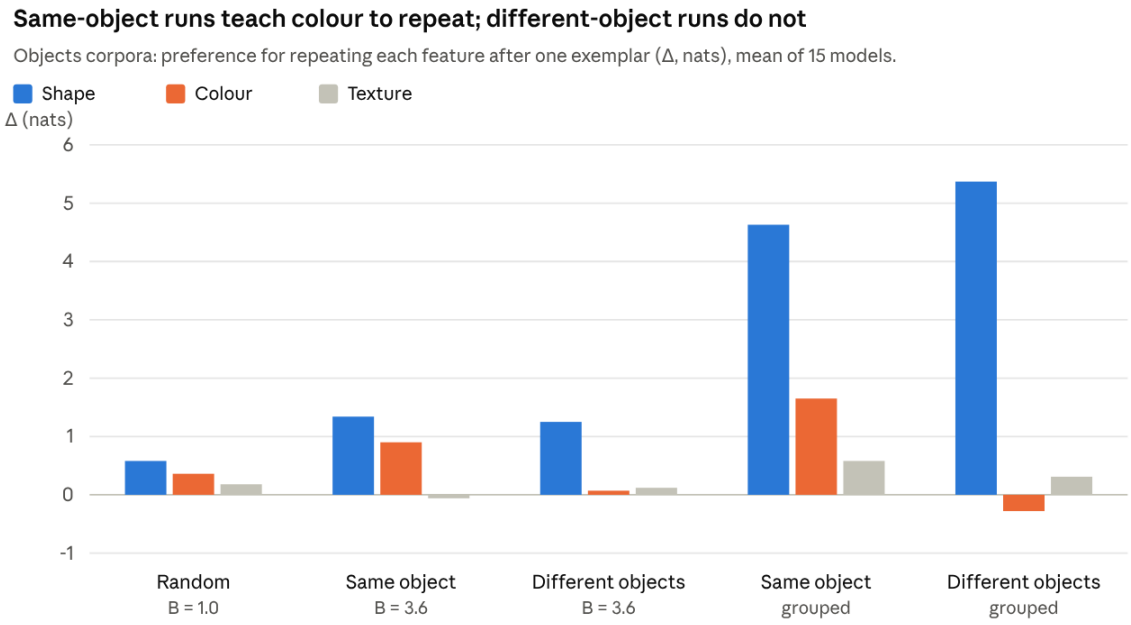


Figure E1: Preference for repeating each feature after one exemplar (Δ, nats; mean of 15 models per condition), in fixed-file models trained on the Objects corpora. Same-object runs raise the preference for repeating colour along with shape; different-object runs raise only shape.

F Trained nouns

All Study 1 models except those trained on the further content controls (Appendix D), 420 in all, were also scored on 960 items whose nouns occurred in training, in the format of Section 3: 16 context nouns, 10 draws of features, three probed dimensions and two continuation nouns (the same noun or another trained noun). No shape that is the fixed feature of either noun appears in an item.

Table F1: Trained nouns, selected conditions (all conditions in the repository). Rows marked fresh are fresh-text models; the rest are fixed-file models. Except after shape reassignment, the index on trained nouns is closer to zero than on novel nouns. Indices in nats; "models above zero" and p (two-sided one-sample t over 15 models) refer to the noun-specific index. For grouped Regular fixed-file models the registered one-sided test gives p = .008.

Condition	Index, same noun	Noun-specific index	Models above zero	p (two-sided)
Regular, random, fresh	−0.02	−0.09	7 of 15	.59
Regular, B 3.5, fresh	−0.13	−0.15	7	.19
Regular, grouped, fresh	2.00	1.04	10	.009
No stable feature, grouped, fresh	−3.75	−2.60	0	$< .001$
Objects, same-object, grouped, fresh	−0.55	0.05	8	.74
Objects, different-object, grouped, fresh	0.03	0.44	13	$< .001$
Reassigned, random, fresh	0.14	−0.07	5	.51
Reassigned, grouped, fresh	11.65	11.24	15	$< .001$
Regular, random	0.07	0.01	9	.97
Regular, B 3.6	0.36	0.12	10	.44
Regular, B 7	0.49	0.18	8	.29
Regular, B 12	1.07	0.06	8	.64
Regular, grouped	1.03	0.30	10	.015
Objects, same-object, grouped	−1.53	−1.06	0	$< .001$
Objects, different-object, grouped	0.33	0.14	10	.26

The comparison of novel and trained nouns in Table L2 is exploratory and was added after the registrations; seed-level intervals for every fresh-text condition are produced by `scripts/recut_noun_store_check.py`.

G Materials for Studies 2 and 3

Object nouns (Study 2). ball, balloon, block, book, crayon, doll, puzzle, toy; airplane, bicycle, boat, bus, car, motorcycle, stroller, tractor, train, truck; apple, banana, carrot, cookie, cracker, egg, grape, raisin, sandwich; boot, button, coat, diaper, hat, jacket, mitten, necklace, pajamas, pants, shirt, shoe, sock, sweater, zipper; basket, blanket, bottle, bowl, box, broom, clock, cup, dish, fork, hammer, key, lamp, pencil, pillow, plate, purse, scissors, spoon, telephone, toothbrush, towel; bed, chair, couch, crib, table, television.

Coding rules (Study 2). Each pair was shown with the utterance before it, the stretch between the two mentions and the utterance after, without its source. *Same*: both mentions refer to the same physical object. *Different*: two or more distinct objects of the kind (another, other, numbers, plurals that count objects, descriptions that cannot fit one object). *Generic*: at least one mention refers to the kind in general. *Unclear*: the words do not settle it, including labels not used for a physical object. The second coder coded a random 25% of pairs (seed 2031). The full rubric and the second coder's prompt are in the repository.

Feature words (Study 3). Word list 1: shape *pointed, oval, circular, curved, rectangular, triangular, spherical, cylindrical*; colour *grey, purple, amber, violet, scarlet, crimson, maroon, turquoise*; texture *smooth, shiny, sticky, fuzzy, fluffy, furry, glossy, silky*. Word list 2: shape *oblong, hexagonal, conical, tapered, elliptical, tubular, cubic, rounded*; colour *magenta, lilac, teal, beige, indigo, lavender, ivory, coral*; texture *velvety, waxy, prickly, bumpy, gritty, coarse, slippery, rough*. Mean Zipf frequencies for list 1 are 3.75 (shape), 3.82 (colour) and 3.70 (texture).

Nouns (Study 3). Original items: snerg, twick, vulf, wazz, glorp, yeff, zull, kleb. Added sets: these and brell, sprock, fendle, plim, skeb, tulver, grisp, vosh.

Item counts (Study 3). Original items: 576 one-exemplar items (8 nouns, 12 draws, 3 dimensions, 2 continuation nouns) and 2,304 with two or four exemplars sharing either shape or colour, 2,880 in all. Each added set has 1,152 one-exemplar items.

Checkpoints (Study 3). Pythia 14M, 70M, 160M, 410M, 1B, 1.4B and 2.8B at steps 0, 4, 16, 64, 256, 1,000, 4,000, 16,000, 64,000 and 143,000 (2.8B without step 64,000); Pythia 6.9B and 12B at steps 64, 1,000, 4,000, 8,000, 12,000, 16,000, 32,000 and 143,000; Pythia 410M to 2.8B also at every 1,000 steps from 2,000 to 15,000 and at 18,000, 20,000, 24,000 and 32,000; nine further training seeds of Pythia 14M, 31M, 70M, 160M and 410M (and the original 31M model) at steps 4,000, 16,000, 64,000 and 143,000; the BabyLM GPT-2 baselines and Qwen2.5 72B and Llama 3.1 70B at their released weights. Exact revisions are listed in the repository.

H Data and code

All code, corpora, test items, model scores, transcript-level counts, hand codes (without transcript excerpts) and analysis outputs, and the final weights and training logs of the 510 trained Study 1 models, will be released on publication; anonymised copies of the code and results are in the supplementary material. A README gives the command that regenerates each table and figure, with MD5 checksums for the fixed inputs. The pretrained models are public at the revisions listed. CHILDES and CABank transcripts are available from TalkBank under its terms and are not redistributed. Study 2 is a secondary analysis of these archived, de-identified transcripts and of licensed telephone corpora, used under their terms; no data were collected from people for this paper, and no transcript excerpts are reproduced.

I Study 1: corpora, training and further results

The main text summarises the design. This appendix gives it in full, with results not reported there.

I.1 Corpora

Each corpus describes 32 kinds of object, each named by a novel noun (*wug*, *dax*, *blorp* and so on), with 14 exemplars per kind. Each exemplar has a shape, a colour and a texture, each a novel word from a list of ten per dimension, and is described by one sentence in one of five frames:

1. *A wug is a [shape] [colour] [texture] thing.*
2. *Look at the wug. It is [shape] and [colour].*
3. *Here is another wug. This one is [shape].*
4. *The wug I showed you is [shape], but a different [colour].*
5. *That is a wug because it is [shape].*

Shape is named in all five frames, colour in three and texture in one. The frames are the same in every corpus we compare, so they cannot by themselves explain a difference between corpora, although they may interact with what is manipulated. The 448 sentences of a corpus, each used once, make up one pass. Five corpora were generated for each condition, one per seed.

Content. The corpora differ in which feature, if any, stays fixed within a kind (Table eftab:1). In **Regular** corpora every kind keeps one shape across its 14 exemplars, while colour and texture vary. In **Mixed-slot** corpora each kind keeps one word from the shape list, but for about half the kinds that word appears in the texture slot, and colour varies; no single dimension is fixed across all kinds. In **No stable feature** corpora, built from Regular, each exemplar's shape is drawn at random from the list, so nothing is fixed within a kind. In **Colour stable** corpora, also built from Regular, shape is drawn in the same way and each kind keeps one colour instead. Frame 4 is unchanged, so in Colour stable corpora it names the kind's fixed colour after the words *a different*.

Objects. A further set of corpora separates two kinds of nearby repetition. In the Regular corpora, two sentences about a kind always describe different exemplars. In speech, nearby mentions often return to

the same object. The Objects corpora keep the Regular content but give each kind 13 objects and describe each object four times, once in each of frames 1, 2, 3 and 5. Frame 4 is left out because it states that the colour differs. A pass of these corpora has 1,664 sentences.

Order. The same sentences were arranged in several orders. In **random** order every pass is shuffled. In **grouped** order, within each pass, the sentences about each kind are shuffled, cut into runs of four, and the runs are shuffled across kinds (Figure 1). Every sentence occurs exactly as often as in random order. In **graded** orders each run is kept with a set probability and otherwise split into single sentences before shuffling, which gives intermediate levels of clustering. In the Objects corpora, a **same-object run** is the four descriptions of one object, and a **different-object run** is four different objects of one kind, one per frame. A Latin square over objects and frames ensures that every sentence is used once per pass in every order. Same-object and different-object runs were each arranged fully grouped and at an intermediate level.

Reassigned. In every corpus so far, a kind's fixed feature is fixed for the whole corpus, so the noun alone predicts it, and a model can predict the shape in a sentence from its noun without comparing the sentence with the one before. In Reassigned corpora, built from Regular, each kind receives a new shape, drawn at random from the ten, in every pass, and keeps it in all of its sentences in that pass. The next sentence about the same kind still repeats the shape, but the noun no longer says which shape it will be. These corpora were arranged in random and grouped order.

Fresh and fixed text. A model trained many times over on one short file can memorise it. Our main models were therefore trained on fresh text: 7,500 passes of a corpus (2,100 for the Objects corpora), each arranged with its own random draw by the rule of its order, 34 to 35 million tokens in all. Training read this text in non-overlapping 128-token windows, so no window was seen twice. A second set of models was trained in the common way, on one fixed file of 11,648 sentences (26 passes, or 7 for the Objects corpora; about 120,000 tokens), with windows starting at random positions, so that each position was seen 160 to 290 times. These fixed-file models memorised their files. Their loss was 0.08 to 0.11 nats per token on their own file but 0.66 to 1.26 on a new arrangement of the same sentences, while fresh-text models scored 0.28 to 0.55 on both (Table C2). Table I1 shows which conditions were trained on fresh text and which on a fixed file; where both exist, we report fresh text first. The fresh graded order, built by the same rule as the fixed one at B = 3.6, measured B = 3.5.

Clustering. We measure clustering with one statistic, B, here and in Study 2. Take the sequence of mentions of object nouns in a text, and every pair of consecutive mentions that are no more than 100 words apart. P is the share of these pairs that name the same kind. P_0 is the same share after the sentences (or utterances) of each document are shuffled, averaged over shuffles. B = P/P_0, pooled over documents. For example, in *Look at the dog. The dog is big. Where is the cup? Here is another dog.*, the consecutive pairs are dog–dog, dog–cup and cup–dog, so P = 1/3. B = 1 means no clustering beyond what the content predicts. Random-order Regular corpora have B = 0.9, slightly below 1, because each pass is shuffled separately: within a pass, a sentence's kind has 13 other members among 447 sentences, against 363 among 11,647 when the whole corpus is shuffled. The graded levels (Table eftab:1) were chosen after the clustering of natural input had been measured (Study 2): 1.4 and 3.6 match B in written text (1.43) and in speech to children (3.57). They were fixed from the corpora alone before any graded-order model was trained.

I.2 Models and training

We trained decoder-only transformers of three sizes: 3.4 million parameters (4 layers, 4 heads, width 256), 11.0 million (6 layers, 8 heads, width 384) and 25.6 million (8 layers, 8 heads, width 512). They were trained for 5,000, 8,000 and 10,000 steps with batches of 32, 32 and 16, peak learning rates of 3×10^{-4}, 2×10^{-4} and 1×10^{-4}, AdamW (betas 0.9 and 0.95, weight decay 0.1), linear warm-up, cosine decay and dropout of 0.1. Each training example is a 128-token window, which spans several sentences, so attention crosses sentence boundaries. The tokeniser for the Regular corpora has 351 entries (314 to 339 for the other corpus families), and every artificial word is a single token. A seed sets the corpus, the initial weights and the batch order (Table I1 gives the models per condition). For the window test, models were trained with 512-token windows and a quarter of the batch size, so that every step saw the same number

Table I1: Study 1 corpora. Every corpus describes 32 kinds. Fresh: 7,500 passes (2,100 for Objects), each arranged anew and read once in non-overlapping 128-token windows. Fixed: one file of 11,648 sentences (10,304 to 12,992 for three further controls: No kind label, Paraphrased and Weak label), read in 128-token windows at random positions unless marked. Each order of each corpus was trained at three sizes with five seeds (15 models). The further controls are reported in Appendix D (Table D1).

Corpus	What stays fixed within a kind	Orders (measured B)	Text	Models
Regular	Shape. Colour and texture vary.	Random (0.9); graded (3.5); grouped (23)	Fresh	45
No stable feature	Nothing	Grouped (23)	Fresh	15
Objects	Shape; each object described four times	Same-object runs (24); different-object runs (24)	Fresh	30
Reassigned	Shape within a pass; a new shape for each kind in every pass	Random (0.9); grouped (23)	Fresh	30
Regular	Shape	Random (0.9); graded (1.4, 3.6, 7, 12); grouped (23)	Fixed	90
Regular	Shape	Random (0.9); graded (3.6)	Fixed, 512-token window	30
Mixed-slot	One shape-list word, in the shape slot for about half the kinds and the texture slot for the rest	Random; grouped	Fixed	30
No stable feature	Nothing	Random; grouped; random with a 512-token window	Fixed	45
Colour stable	Colour. Shape varies.	Random; grouped	Fixed	30
Objects	Shape; each object described four times	Random (1.0); same-object runs (3.6, 24); different-object runs (3.6, 24)	Fixed	75
Further controls: no kind label, weak label, paraphrased, frequency matched, added noise, feature swap	As Regular, with one change each	Random	Fixed	90
Total				510

of tokens. All models have separate input and output embeddings, and the parameter counts include both; the model code ties the two, but the tie was lost on the training hardware (Appendix A). Table C1 lists every setting.

I.3 Measuring the expectation

Eight novel nouns that never occur in training (*snerg*, *twick* and so on) are used in test items. They are in the vocabulary, so each is one token, but their input embeddings were never trained; weight decay shrinks them by less than 10% over training, so they remain distinct random vectors. With the intended tie, the same vectors would have served as the novel nouns' output weights and would have changed in training; we cannot say whether the models would then have tracked these nouns. Novel nouns keep learned knowledge about a noun out of the test, but the test then depends on the model registering that the same untrained token recurs. We therefore also test with nouns from training, whose embeddings are trained (Appendix F). Each item gives the model one sentence in frame 1 describing an exemplar, followed by two continuations that differ in one word:

> Context: *A snerg is a krannish pulek skorli thing.* Repeat: *A snerg is a* ***krannish*** *temil tramel thing.* New: *A snerg is a* ***tova*** *temil tramel thing.*

In this item the probed dimension is shape. The repeat continuation keeps the context's shape, the new one does not, and both give new values for the other two dimensions. The model scores each continuation by the sum of the log probabilities of its tokens given the context, so the difference, Δ, includes the

changed word and its effect on the words after it. Δ is the model's preference for repeating the probed dimension, in nats. Trained models generally prefer words they have just seen, so Δ alone mixes the dimension with copying. Our index is the difference between dimensions:

$$\text{index} = \Delta_{\text{shape}} - \Delta_{\text{colour}}$$

Copying that affects shape and colour equally cancels. Copying that differs between dimensions does not, which is one reason every conclusion rests on comparisons between corpora tested in the same way, and why we report Δ for each dimension as well. The shape-over-texture index is defined in the same way. With one exemplar, nothing in the context shows which dimension is fixed, so differences between trained conditions come from training.

Every item is also given with a different novel noun in the continuations (*A yeff is a krannish temil tramel thing*). The **noun-specific index** is the index after the same noun minus the index after a different noun. A kind-tied expectation scores above zero on it; a noun-blind one scores zero.

With 25 draws of features per noun, three probed dimensions and two continuation nouns there are 1,200 one-exemplar items. A further 2,400 items with two or four exemplars test the use of evidence in the context (Table E2). Untrained models of the same architecture, with the Regular tokeniser, give an index of 0.02 nats (SD 0.07).

I.4 Statistics

Each model gives one index, its mean over items. Models of the three sizes trained with the same seed share a corpus, so the five seeds are the units of replication. We average the three sizes within each seed and compare conditions by paired t tests over the five seed means (4 degrees of freedom); the differences and 95% intervals given in the text are these seed-level estimates. Registered contrasts were tested one-sided, with Holm (1979) correction within each registered family. For fresh-text models the registered test was this seed-level test. For fixed-file models it was a linear mixed model (index $\sim$ condition + size, with a random intercept for seed) or, for some contrasts, a Welch t test over models; Table E1 gives the seed-level results beside them. Single conditions were compared with zero by one-sample t tests. Results reported with a Holm-corrected p, or called registered, were registered; all others are exploratory. We registered 0.2 nats as the smallest effect of interest and tested equivalence to zero against it with two one-sided tests (Lakens, 2017). This bound is small next to the effects reported here, so the equivalence tests are strict.

I.5 Further results

Mixed-slot corpora, trained as fixed files, keep a fixed word for every kind but no single fixed dimension. In grouped order these models preferred shape over colour (2.12), as their corpus supports, since colour is never fixed there, but showed no reliable preference between shape and texture (0.46), each of which is fixed for about half the kinds. Grouped Regular models were above them on both indices (differences 4.37 [1.98, 6.76] and 4.89 [2.69, 7.10], Holm $p < .001$). In random order, Regular models were above Mixed-slot models by the registered test (0.47 against −0.25, Holm p = .032), but not reliably at the seed level (3 of 5 seeds; p = .11).

Across the six orders of the fixed-file Regular corpora, the index rose with clustering at every step (Table E4, Figure I1). The registered test, a linear trend in the natural log of B, was supported (slope 1.63 nats per log unit, SE 0.10, Holm $p < .001$; seed level, 1.63 [1.46, 1.80]). The rise was steeper at high clustering: about 0.6 to 1.0 nats per log unit up to B = 7, and 2.7 and 4.6 nats per log unit over the two highest steps. At B = 3.6, fixed-file models were above random order (1.43 against 0.47; difference 0.96 [0.63, 1.28], Holm $p < .001$), as were fixed-file Objects models with different-object runs at the same level (1.18 against 0.22; difference 0.96 [0.37, 1.54], Holm $p < .001$). Fresh-text models agree with this curve at B = 3.5, but lie below it in random order and above it in grouped order (Figure I1).

In fixed-file models, under our matched-tokens recipe, a longer window did not raise the index (with tokens per step held fixed, the 512-token models also saw a quarter as many sequences per step, so the result applies to this recipe, not to window length alone; Table C1). With 512-token windows, which hold

about four times as many sentences, random-order Regular models gave 0.50 nats against 0.47 with 128 tokens (difference 0.03 [−0.44, 0.50], Holm p = .87), and the registered equivalence test did not show equivalence either (p = .15). At B = 3.6, the longer window gave less, 0.80 against 1.43 (difference −0.63 [−0.90, −0.37]; two-sided seed-level p = .003). With the longer window, Regular models were still above No stable feature models (0.50 against 0.10, Holm p = .045; seed level, 4 of 5 seeds, p = .032).

At matched clustering, runs of different objects gave a larger index than runs re-describing one object: 7.60 against 5.87 nats in grouped order (difference 1.72 [0.54, 2.90], Holm p = .008; higher in all five seeds). Fixed-file models showed the same (5.65 against 2.99; difference 2.66 [1.57, 3.75], Holm p $<$ .001), and also at B = 3.6 (1.18 against 0.45; difference 0.73 [0.15, 1.32], Holm p $<$ .001). Their same-object runs at B = 3.6 were neither reliably different from random order (0.45 against 0.22; difference 0.22 [−0.30, 0.74]) nor equivalent to it (registered secondary test).

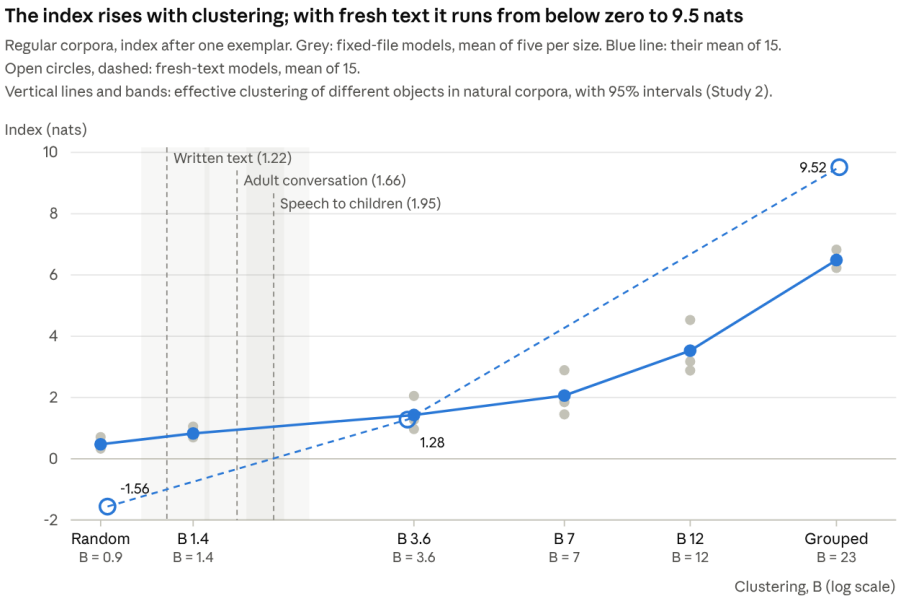


Figure I1: Index against clustering B in Regular corpora, unseen test tokens. Grey points and solid line: fixed-file models (mean of five per size; line, mean of 15). Open circles and dashed line: fresh-text models (mean of 15). Bands: effective clustering of different objects in natural corpora with 95% intervals (Appendix J); the overlay does not predict what a model trained on them would learn. With typical test tokens or trained nouns the fresh-text effect at $B = 3.5$ disappears (Appendix L).

J Study 2: clustering of natural input

Study 2 measures, with the statistic B of Study 1, how clustered mentions of object kinds are in three kinds of natural input: speech recorded around young children, conversation between adults, and the written text language models are trained on. It is descriptive. B is a shared statistic, not a calibrated dose: natural corpora differ from the Study 1 corpora in many ways besides clustering.

J.1 Method

Corpora. We used the North American English collection of CHILDES (MacWhinney, 2000), downloaded in full from TalkBank, kept every transcript whose target child was under 36 months, and took every utterance not spoken by that child. Most of this speech is addressed to the child, but some is addressed to others present; for brevity we call it speech to children. For adult conversation we used CABank: the Santa Barbara Corpus of Spoken American English (Du Bois and Englebretson, 2004), CallHome English (Canavan et al., 1997) and CallFriend English (Canavan and Zipperlen, 1996a,b). All but the Santa Barbara Corpus are telephone calls, in which the speakers share no objects in view. For written text we used the first 10,000 documents of the Pile (Gao et al., 2020), the training data of the Pythia

models in Study 3, split into sentences. Documents of fewer than 400 words or with no mention of a target noun were excluded, leaving 2,116 transcripts from 35 CHILDES corpora, 245 CABank transcripts from four sources and 2,474 Pile documents from 21 subsets.

Kinds. We counted mentions of 70 count nouns from the MacArthur-Bates Communicative Development Inventory (Fenson et al., 2007) that name solid, inanimate objects, such as *ball*, *cup*, *shoe* and *truck*, the kinds for which children show the shape bias. Singular and plural count as one kind, and words often used as verbs or adjectives in speech to children, such as *brush* and *orange*, were left out. Because the nouns come from a list of words young children know, they are more frequent in speech to children than in the other corpora, by design.

Measure. B was computed as in Study 1, with the utterances or sentences of each document shuffled for P_0, within a window of 100 words, about the length of the 128-token training window of Study 1. Intervals resample sources (CHILDES corpora, CABank sources or Pile subsets) and then documents within them.

What repeated labels refer to. B counts repeated labels. It cannot tell two balls from one ball named twice, and Study 1 showed that the difference matters. One author coded 500 random same-label pairs (300 from speech to children, 100 from each of the others), with source labels withheld, as referring to the same object, to different objects of the kind, to the kind in general, or unclear; source could sometimes be guessed from the text. As a model-assisted consistency check, an instance of Claude (Anthropic), given only the coding rules and the pairs, coded a random quarter of them without access to the first coder's codes; the prompt and the codes are archived. Agreement on 124 pairs was 84% (Cohen's κ = 0.78). This is not a measure of agreement between human coders.

Effective clustering (exploratory). As a rough projection onto the Study 1 scale, we discounted each corpus's excess clustering by the share of coded pairs that referred to different objects, d, out of those that referred to particular objects: B_{eff} = 1 + (B − 1)d. A strict version takes d out of all coded pairs. Intervals resample sources and documents for B and coded pairs for d. B_{eff} assumes that only different-object pairs inform the learner. Study 1 shows that same-object repetition is not neutral, so B_{eff} is a heuristic, not a corrected estimate.

J.2 Results

Object labels were clustered in every corpus, most of all in speech to children (Table J1). In speech to children B was well above 1 (registered test, Holm p $<$ .001), above written text (difference 2.14 [1.55, 2.91], Holm p $<$ .001) and above adult conversation (difference 1.24 [0.51, 2.03]; registered two-sided test, p $<$ .001). The order of the three held in three registered secondary analyses: when each transcript was weighted equally, when every common noun was counted instead of our 70, and when documents were shuffled only within 500-word blocks, which keeps changes of topic. That last check lowered B in speech to children from 3.57 to 2.08, so part of the raw clustering reflects topic.

Table J1: Clustering of mentions of 70 object nouns within 100 words. Intervals resample sources and then documents, and for B_{eff} also coded pairs. B_{eff} is exploratory.

Measure	Speech to children	Adult conversation	Written text
B [95% CI]	3.57 [3.13, 4.30]	2.33 [2.02, 2.93]	1.43 [1.22, 1.86]
B, each transcript weighted equally	2.91	2.27	1.31
B, shuffling only within 500-word blocks	2.08	1.28	1.16
B, all common nouns	9.65	5.73	2.07
Same-label pairs per 1,000 words	10.1	0.9	1.0
Coded pairs: different objects	27% (82 of 300)	25% (25 of 100)	13% (13 of 100)
Coded pairs: same object	46% (139)	25% (25)	12% (12)
Coded pairs: kind in general or unclear	26%	50%	75%
B_{eff} [95% CI]	1.95 [1.73, 2.28]	1.66 [1.44, 2.04]	1.22 [1.09, 1.47]
B_{eff}, strict	1.70 [1.53, 1.96]	1.33 [1.21, 1.54]	1.06 [1.02, 1.13]

Nearly half of the repeated labels in speech to children referred again to the same object, and about a quarter brought a different object, a share similar to adult conversation. Most repeated labels in written text did not refer to a physical object, or referred to the kind in general. Effective clustering of different

objects was therefore lower than B in every corpus. Speech to children was above written text on this measure (difference 0.73 [0.41, 1.09]) but not reliably above adult conversation (0.29 [−0.14, 0.69]).

Placed on the Study 1 scale by effective clustering, as registered, written text lies between random order and the graded order at B = 1.4, and adult conversation and speech to children lie between the graded orders at B = 1.4 and B = 3.6 (Figure I1). Over that range the index of fixed-file models rose only from 0.47 to 1.43 nats; fresh-text models were trained only near its ends, where they gave −1.56 in random order and 1.28 at B = 3.5. This does not say what a model trained on each corpus would learn.

The corpora differ more in how often such repetitions occur. Speech to children contained 10.1 same-label pairs per 1,000 words, against about 1 in the other two. Combining these rates with the coded shares, it contained about 2.8 nearby repetitions per 1,000 words that bring a different object of a kind, against about 0.2 in adult conversation and 0.1 in written text (exploratory). Part of this difference comes from our choice of nouns. B is a ratio and does not capture density, which Study 1 held fixed.

K Study 3: pretrained models

Study 3 asks whether models trained on large written corpora show the expectation, how it depends on size and training, and whether they tie it to the noun. It is a descriptive comparison: the arrangement of their training data is not manipulated.

K.1 Method

Items. The items follow the format of Study 1 in English. Each feature gets its own sentence, so English adjective order cannot favour one dimension, and the order of the three sentences is counterbalanced:

> Context: *Here is a snerg. It is cylindrical. It is purple. It is smooth.* Repeat: *Here is another snerg. It is* ***cylindrical****. It is turquoise. It is fluffy.* New: *Here is another snerg. It is* ***circular****. It is turquoise. It is fluffy.*

Each dimension uses eight English words of similar frequency, and the nouns are novel words spelled with ordinary subword pieces (Appendix G). In the original items the same-noun continuation began *Here is another snerg* and the different-noun continuation *Here is a zull*, so the noun-specific index compared the noun and the word *another* together. Four further item sets with 16 nouns keep the wording of the two continuations the same: *a* in both (the registered primary set), *another* in both, a second frame, and a second word list.

Models. The Pythia models (Biderman et al., 2023; van der Wal et al., 2025) range from 14 million to 12 billion parameters and were trained on the same 300 billion tokens of the Pile in the same order. We scored their final checkpoints and checkpoints across training, and nine further training seeds of the five smallest sizes (van der Wal et al., 2025). We also scored the two GPT-2 baselines of the BabyLM Challenge (124 million parameters, trained on 10 and 100 million words; Charpentier et al., 2025), and Qwen2.5 72B (Qwen Team, 2024) and Llama 3.1 70B (Grattafiori et al., 2024) as descriptive reference points. Appendix G lists every checkpoint.

Tests. Trends with size are Spearman correlations across Pythia sizes; trends over training are tested by permuting checkpoints within size; intervals resample nouns. The pooled noun-specific test uses a mixed model with crossed random effects for nouns, items and feature words.

K.2 Results

The index rose with size but not steadily over training. At the end of training the shape-over-colour index rose with Pythia size (Spearman ρ = .92, Holm p = .003), from −2.44 nats at 14 million parameters to 1.17 at 12 billion (Figure K1). This raw index includes effects of the particular words. Untrained Pythia models give about 0.5 nats, because in some colour pairs the repeated and the new word take different numbers of tokens, and an untrained model pays roughly the same cost for every token. On the original items whose two words take the same number of tokens (all shape pairs and a quarter of the colour pairs; exploratory), untrained models gave −0.32 to 0.31 nats, the trend with size held (ρ = .93), and every Pythia model from 1B upwards gave an index above zero, from 1.00 [0.04, 1.82] at 1B to 1.62 [0.86, 2.19] at 12B.

The trend with size held on every item set, including the second word list (ρ = .83 to .95; exploratory). The index did not rise steadily over training (mean within-size ρ = −.34). At every size, training first lowered it, to about −2 nats between steps 1,000 and 4,000, much as in the random-order models of Study 1, and only the larger models recovered. The BabyLM baselines did not show a reliable index (10 million words, 0.47 [−1.05, 1.93]; 100 million words, 0.37 [−0.04, 0.73]; Holm p = .52 and .12). On the items whose two words take the same number of tokens (exploratory), the 100-million-word baseline gave 1.16 [0.52, 1.82]. Pythia 160M, of similar size, gave a negative index (−1.22 [−1.66, −0.75]) even after 300 billion tokens, but the two families differ in more than their data, so this comparison does not separate data from size.

Weak, wording-dependent evidence of noun binding. On the original items the noun-specific index was positive in every Pythia model from 410 million parameters, but those items confounded the noun with *another*. With *a* in both continuations, the registered tests were not supported: the index did not rise with size (ρ = −.42, Holm p = 1.00), and pooled over Pythia 410M to 12B it was not above zero (−0.15 nats, SE 0.12, Holm p = 1.00). It was higher at 410M than at 160M across ten training seeds of each (0.15 against −0.15 nats, Holm p = .012), with 7 of the 10 seeds at 410M positive. With *another* in both continuations (a registered secondary analysis), it was above zero in four of the six Pythia models from 410M (0.28, 0.77, 0.30 and 0.17 nats at 410M, 1B, 2.8B and 6.9B), but not at 1.4B or 12B, and it was small and mixed in sign on the second frame and the second word list. Of the two reference models of about 70 billion parameters, Qwen2.5 gave a positive index with *a* in both continuations (0.56 [0.28, 0.84]) and Llama 3.1 did not (−0.16 [−0.36, 0.06]).

The components show why the wording matters. After *another*, which marks a new member of the kind, the larger models expected shape to repeat and colour to change more after the same noun than after a different one, and *another* also raised the raw index, to 0.49 to 1.08 nats in Pythia 1B to 12B (−0.34 to 0.46 with a in both). After *a*, the same noun may name the same object again, and then every feature should repeat: colour repeated more after the same noun, and the noun-specific index fell.

Texture and context. No Pythia, Qwen or Llama model preferred shape over texture, so what the large models show is a shape-over-colour expectation, not a general shape bias (image classifiers trained on ImageNet likewise favour texture over shape; Geirhos et al., 2019). The BabyLM baseline trained on 10 million words did prefer shape over texture (1.17 [0.24, 2.34]). The two largest Pythia models tied texture to the noun more than shape. When two or four exemplars in the context shared a shape, every trained model followed the context, strongly so from 410 million parameters. What developed with size was the expectation after one exemplar, not the use of evidence in the context.

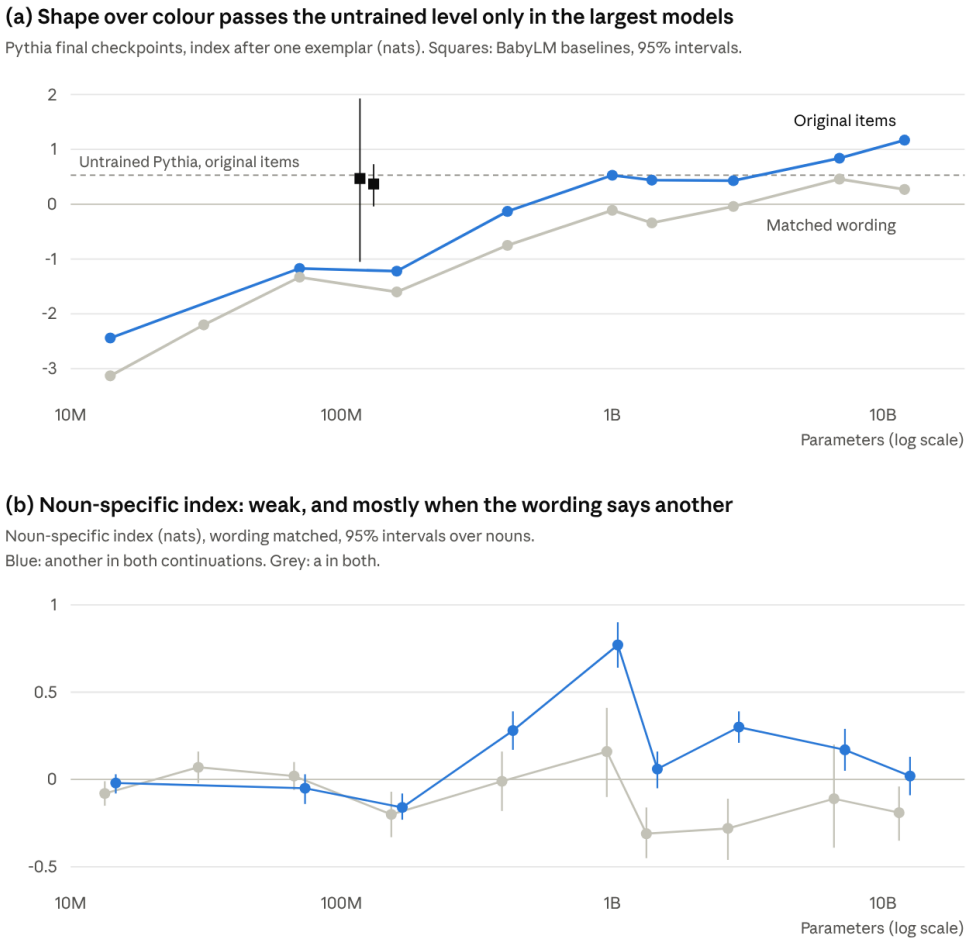


Figure K1: Pretrained models. (a) Shape-over-colour index (nats) at the end of training by Pythia size, on the original items and with matched wording (*a* in both continuations). Squares: BabyLM baselines with 95% intervals over nouns. Dashed line: untrained Pythia models on the original items (about 0.5 nats, from differences in token counts; Appendix K). (b) Noun-specific index with matched wording, with *a* or *another* in both continuations, with 95% intervals over nouns. In both panels, the 31M model was scored only with *a* in both continuations.

L Test-noun representation and changed-word scoring

These analyses were added after the registrations and are exploratory. They rescore the 120 fresh-text models on the one-exemplar items (`scripts/recut_rescore.py`, `scripts/recut_rescore_analysis.py`). **Typical tokens**: each novel noun's input embedding is replaced by the mean input embedding of the 32 training nouns plus $r\,u_j$, where r is the mean distance of the training-noun embeddings from that mean and the u_j are orthonormal random directions (fixed seed), so the eight tokens look like ordinary nouns but stay distinct; output weights are unchanged. **Changed-word** Δ: the difference in log probability of the one word that differs between the continuations, given the identical prefix. With the original embeddings and whole-continuation Δ, the rescoring reproduced the reported scores (144,000 item scores; largest difference 2×10^{-5} nats). Trained-noun values are from the registered trained-noun items (Appendix F; `scripts/recut_noun_store_check.py`).

Mean input-embedding norms: trained nouns 0.55, unseen tokens 0.36, typical tokens 0.55.

Table L1: Index and noun-specific index under four scorings (nats; mean of seed means). Original: unseen tokens with untrained embeddings, as reported; typical: the same tokens with typical embeddings; full: whole-continuation Δ, as reported; word: changed word only.

	Index				Noun-specific index			
Condition	orig./full	typ./full	orig./word	typ./word	orig./full	typ./full	orig./word	typ./word
Regular, random	−1.56	−1.54	−1.26	−1.50	−0.05	−0.31	0.01	−0.20
Regular, graded (3.5)	1.28	−1.41	1.45	−1.25	−0.03	−0.15	0.03	−0.04
Regular, grouped	9.52	1.46	9.65	1.38	0.12	0.00	0.12	0.01
No stable feature, grouped	−3.98	−3.51	−3.74	−3.24	−0.35	−0.26	−0.34	−0.21
Objects, same-object runs	5.87	−1.85	6.06	−1.87	−0.07	0.14	−0.06	0.06
Objects, different-object runs	7.60	−1.25	7.59	−1.33	−0.01	0.19	−0.01	0.11
Reassigned, random	0.21	0.18	0.32	0.27	0.02	−0.10	0.03	−0.01
Reassigned, grouped	9.80	3.87	9.94	3.98	0.37	0.70	0.37	0.74

Table L2: Preference for repeating the context's shape and colour after one exemplar, same noun, for the three kinds of test noun (nats; whole-continuation Δ).

	Δ shape			Δ colour		
Condition	Unseen	Typical	Trained	Unseen	Typical	Trained
Regular, random	−1.55	−1.52	0.03	0.01	0.02	0.05
Regular, graded (3.5)	1.18	−1.52	0.06	−0.10	−0.12	0.19
Regular, grouped	6.88	−0.72	0.06	−2.64	−2.18	−1.94
No stable feature, grouped	−3.88	−3.51	−3.58	0.10	0.00	0.17
Objects, same-object runs	6.87	−1.17	−0.09	0.99	0.68	0.46
Objects, different-object runs	8.05	−1.14	0.04	0.46	0.11	0.01
Reassigned, random	0.22	0.24	0.24	0.02	0.06	0.10
Reassigned, grouped	6.81	1.17	10.01	−2.99	−2.70	−1.65

Table L3: Noun-familiarity curve: unseen tokens' input embeddings blended from their untrained values (0) to typical embeddings (1). Index and Δ shape, same noun (nats; mean of seed means); slope per unit blend with 95% interval over five seeds.

Condition	Measure	0	0.25	0.5	0.75	1	Slope
Regular, random	Index	−1.56	−1.83	−1.93	−1.79	−1.54	0.03 [−0.22, 0.28]
Regular, random	Δ shape	−1.55	−1.83	−1.92	−1.78	−1.52	0.05 [−0.26, 0.35]
Regular, grouped	Index	9.52	6.70	4.32	2.10	1.46	−8.28 [−9.87, −6.70]
Regular, grouped	Δ shape	6.88	4.27	2.08	−0.08	−0.72	−7.82 [−9.59, −6.04]
Objects, same-object runs	Index	5.87	3.71	1.08	−1.41	−1.85	−8.23 [−10.05, −6.40]
Objects, same-object runs	Δ shape	6.87	4.59	1.90	−0.65	−1.17	−8.52 [−10.22, −6.83]
Objects, different-object runs	Index	7.60	4.53	1.87	−0.82	−1.25	−9.22 [−9.97, −8.46]
Objects, different-object runs	Δ shape	8.05	4.85	2.08	−0.69	−1.14	−9.57 [−10.14, −9.00]
Reassigned, grouped	Index	9.80	8.03	5.34	3.87	3.87	−6.41 [−8.74, −4.08]
Reassigned, grouped	Δ shape	6.81	4.83	2.24	0.97	1.17	−6.06 [−7.28, −4.84]

M The familiarity study (Amendment 9)

Registration. The design, hypotheses and analysis were registered before any corpus was built (Amendment 9). An addendum, written the same day before any corpus was built, fixed details the registration left open (36 context draws per noun; uniform frame proportions; exposures placed at evenly spaced read positions from 5% to 95% of training, after warm-up for every size; random in-window position; the 40 nonce nouns; how the pilot fixes the number of nouns per cell; how a hypothesis counts as supported; the gate for H35; and that heads without prefix matching are described as copying heads). A pilot (one seed, 3.4M models, never used in a test) set three nouns per cell; with the registered rule the expected power for a 1-nat difference was 0.57, below the planned 0.8, which we report as a shortfall. Corpus checksums were recorded before training and matched on the training machine; a checker verified every exposure, window and frame from the tokens. All models were trained on one machine.

Design. In every model, 40 nonce nouns fill 13 cells (three per cell, four unseen): fixed shape,

uninformative shape (a new shape in every sentence, drawn from the corpus's shape distribution) and shape-absent (a sixth frame, *The noun is colour and texture*, also used by trained kinds in one sentence in fifteen), each at 10, 100, 1,000 and 10,000 exposures, plus unseen. Nouns rotate over cells across ten seeds. Each familiarity sentence lies wholly inside one 128-token window, at most one per window, between runs in grouped corpora. Base corpora: Regular grouped, Regular random, Reassigned grouped; sizes 3.4M, 11.0M and 25.6M; ten seeds; 90 models with untied embeddings. Measures are read from the full log-softmax over the ten shape words at the position that predicts the shape in *A noun is a*: carry $= L(s) - \bar{L}$ after a context sentence with shape s; stored preference $= L(s^*) - \bar{L}$; knowledge $= L(s^*)$ minus the mean over the other nine shapes after an unrelated sentence, where $\bar{L}$ is the mean over shapes other than s and s^*.

Table M1: Registered tests of the familiarity study (Regular grouped models unless stated; seed-level estimates over ten seeds, sizes averaged within seed; nats, proportions for H33a, attention probability for H35). Holm p is adjusted over H30 to H34; H35 was not entered because H33(a) failed. Sizes: number of the three model sizes with the registered direction.

	Test	Estimate	95% CI	Holm p	Sizes	Verdict
M1	Knowledge at 1,000 exposures, fixed shape (each corpus)	9.53; 8.61; 8.26				Passed
H30	Carry falls faster with exposure for fixed-shape than shape-absent nouns (slope difference)	0.16	[0.01, 0.30]	1.000	0	Not supported
H31	Shape-absent (1k and 10k) minus unseen with typical embedding (two-sided; TOST ± 1)	0.33	[0.13, 0.52]	.013	3	Different, and equivalent
H32	Stored preference rises with exposure, fixed shape (slope)	2.97	[2.79, 3.16]	$<.001$	3	Supported
H33a	Ablating candidates removes more than half of unseen carry (proportion)	0.22	[0.07, 0.36]	1.000	2	Not supported
H33b	Random heads minus candidates, unseen carry after ablation	0.64	[−0.43, 1.71]	1.000	2	Not supported
H34	Reassigned: trained-kind carry intact minus candidates ablated	7.41	[6.02, 8.80]	$<.001$	3	Supported
H35	Attention slope difference, fixed shape minus shape-absent	0.002	[0.000, 0.005]	–	2	Not tested (gate)
R1	Replication: grouped minus random Regular, unseen carry	4.68	[3.71, 5.65]	$<.001^a$	3	Replicated

[a] Outside the Holm family (replication); unadjusted one-sided p. M1 values: Regular grouped; Regular random; Reassigned grouped (all with slopes above zero, $p < .001$). H31 TOST $p < .001$.

Table M2: Carry (nats) by noun type and exposure, mean of seed values (sizes averaged). Unseen: never in training; typical: the unseen nouns with a typical embedding; trained: the 32 base kinds.

Corpus	Unseen	Typical	Trained	Familiar nouns	10	100	1k	10k
Regular grouped	4.53	−0.33	−0.02	Fixed shape	3.70	0.30	−0.30	−0.37
				Uninformative	3.57	0.16	0.00	−0.01
				Shape-absent	4.47	0.86	0.05	−0.05
Reassigned grouped	6.85	1.57	9.98	Fixed shape	6.40	2.13	0.70	0.60
				Uninformative	6.34	2.53	0.30	0.15
				Shape-absent	6.97	4.89	2.04	1.86
Regular random	−0.15	−0.72	−0.10	Fixed shape	−0.13	−0.13	−0.06	−0.05
				Uninformative	−0.10	−0.03	−0.02	−0.01
				Shape-absent	−0.19	−0.09	−0.07	−0.05

Table M3: Knowledge and stored preference (nats) in Regular grouped models, mean of seed values.

Type	Measure	10	100	1k	10k
Fixed shape	Knowledge	1.59	8.02	9.53	10.61
	Stored	1.41	7.74	9.56	10.71
Uninformative	Knowledge	0.04	−0.03	0.00	0.03
	Stored	0.01	−0.03	−0.01	0.03
Shape-absent	Knowledge	−0.14	−0.17	0.12	0.08
	Stored	0.11	−0.16	0.10	0.06

Table M4: Head analysis by corpus and size (means over ten seeds). Candidates were the k heads with the highest product of prefix-matching and copying scores; in grouped models they were in layer 0 in all but one model (a 3.4M Regular grouped model chose one layer-1 head), and in random-order models a few were in later layers. In the 3.4M and 11.0M models layer 0 has exactly k non-candidate heads, so the random draws repeat one set. Prefix: highest prefix-matching score of any head (uniform attention gives about 0.027). Carry of unseen nouns and of trained kinds intact, with the candidates mean-ablated, and with random heads ablated (mean of 200 or 20 draws). DLA: change in unseen carry predicted by the candidates' direct paths.

				Unseen carry			Trained carry			
Corpus	Size	k	Prefix	Intact	Cand.	Random	Intact	Cand.	Random	DLA
Regular grouped	3.4M	2	0.028	8.10	−0.42	5.26	−0.25	−0.85	−0.71	−0.04
	11.0M	4	0.037	2.98	5.80	1.95	0.08	0.17	−0.40	−0.01
	25.6M	6	0.037	2.50	0.73	0.81	0.12	−0.60	−0.18	−0.01
Reassigned grouped	3.4M	2	0.029	6.48	4.22	3.27	9.59	2.83	3.37	−0.12
	11.0M	4	0.032	7.21	6.56	4.82	10.10	5.02	2.45	−0.03
	25.6M	6	0.031	6.88	0.26	3.26	10.26	−0.11	3.39	−0.01
Regular random	3.4M	2	0.028	−0.24	−0.09	−0.34	−0.08	0.00	−0.11	−0.01
	11.0M	4	0.030	0.04	0.25	0.03	−0.06	0.04	−0.11	−0.01
	25.6M	6	0.030	−0.25	0.00	−0.37	−0.18	−0.10	−0.27	−0.01

Table M5: Attention of the candidate heads from the position that predicts the shape to the context's shape word (mean over candidates; mean of seed values; exploratory).

Corpus	Type	Unseen	10	100	1k	10k
Regular grouped	Fixed shape	0.155	0.144	0.082	0.071	0.071
	Uninformative		0.147	0.088	0.076	0.074
	Shape-absent		0.154	0.097	0.077	0.075
Reassigned grouped	Fixed shape	0.219	0.217	0.108	0.055	0.059
	Uninformative		0.217	0.157	0.071	0.070
	Shape-absent		0.219	0.170	0.115	0.139
Regular random	Fixed shape	0.089	0.088	0.076	0.067	0.067
	Uninformative		0.087	0.082	0.075	0.073
	Shape-absent		0.089	0.084	0.075	0.073